Graph Neural Networks for Molecular Property Prediction

The research task is to predict molecular properties from molecular graphs. We propose a message pa graph neural network with attention over chemical bonds and self-supervised pretraining on unlabeled molecules. The model is evaluated on molecular benchmark datasets with classification and regression targets, compared with fingerprint-based baselines and transformer models over SMILES strings. We a generalization to out-of-distribution scaffolds and the effect of contrastive pretraining objectives.